Interview Practice :→ C Program

① Vowel or Constant

char c → a, e, i, o, u, A, E, I, O, U

#include <ctype>

② Leap Year Check

(n%4 == 0 && n%100 != 0) ||
(n%400 == 0)

3. Number Palindrome :

12321

4. Display All Armstrong Numbers between 1 to (1000)

$153 = 1^3 + 5^3 + 3^3$

$1^1 = 1$

10^3 ④ = 1000

$\log10(num)$ 90

int(3.99) = 3 + 1

<math.h>

n = (int) log10(num) 233 +1

$x^1 = x$

$1^1 = 1$ $2^1 = 2$ $x = 1$

5. Display prime nos b/w a certain range : 1 & itself

[1 (s) – 20 (e)]

[2, 3, 5, 7, 11, 13, 17, 19]

1 X, 2, 3, 4, ... if ×2 (36) ×

n, n/2

(i = 2; i <= n/2 ; i++)

if n%i == 0 → not prime

else → prime

6. Merge two arrays into one single array.

int a1[] = {1, 2, 3} = n1 = 3 (0 1 2)

int a2[] = {4, 5, 6, 7} = n2 = 4 (0 1 2 3)

O/p a3[] = {1, 2, 3, 4, 5, 6, 7}; (0 1 2 3 4 5 6)

index = 0

```
for(int i = 0; i < n1 ; i++)
    a3[index++] = a1[i];
for(int i = 0; i < n2 ; i++)
    arr[n1 + i] = a2[i];
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3 + 0

7. Bubble Sort : n=4, 3 passes (n-1)

